Griffiths Chapter 10: Gauge Theories

A First-Exposure Summary of Sections 10.1–10.4

PHYS 7502 Reading Guide

1 Section 10.1 — Lagrangian Formulation of Classical Particle Mechanics

What is a Lagrangian?

In classical mechanics you already know Newton's second law: $\mathbf{F} = m\mathbf{a}$. The Lagrangian is an *equivalent* way to state the same physics. Define

$$L = T - U$$

where $T = \frac{1}{2}m\dot{q}^2$ is kinetic energy and $U(q)$ is potential energy. The variable q is any **generalized coordinate** (it could be x, or θ, or anything convenient).

Why bother with Lagrangians?

Newton's law requires you to identify forces and pick a coordinate system. The Lagrangian method only asks you to write down energies — T and U — in whatever coordinates are convenient. The equations of motion then follow automatically from one universal recipe: the **Euler–Lagrange equation**.

The Euler–Lagrange Equation (Particle Mechanics)

The Recipe

Given $L(q_i, \dot{q}_i)$, the equation of motion for each coordinate q_i is:

$$\boxed{\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0}$$

Let's verify this reproduces Newton's law in Cartesian coordinates ($q = x$):

$$L = \tfrac{1}{2}m\dot{x}^2 - U(x)$$

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x} \quad \text{(this is the momentum } p\text{)}$$

$$\frac{\partial L}{\partial x} = -\frac{\partial U}{\partial x} = F_x$$

Plugging into the Euler–Lagrange equation:

$$\frac{d}{dt}(m\dot{x}) - F_x = 0 \quad \Longrightarrow \quad m\ddot{x} = F_x \quad \checkmark$$

The pattern to remember

The Euler–Lagrange equation always does two things:

1. Differentiates L w.r.t. the "velocity" $\dot{q}$ to get the **canonical momentum**.
2. Compares that momentum's time derivative against the "force" $\partial L/\partial q$.

This same pattern will carry over directly to field theory.

2 Section 10.2 — Lagrangians in Relativistic Field Theory

Particle → Field: What Changes?

A **particle** is described by coordinates $q_i(t)$ — a finite number of functions of time only.
A **field** is described by $\phi_i(x, y, z, t)$ — functions of *all* spacetime coordinates.

Concept	Particle	Field
Dynamical variable	$q_i(t)$	$\phi_i(\mathbf{x}, t)$
"Velocity"	$\dot{q}_i$	$\partial_\mu \phi_i$ (four–gradient)
Lagrangian	$L(q_i, \dot{q}_i)$	$L(\phi_i, \partial_\mu \phi_i)$ (a *density*)
E–L equation	$\dfrac{d}{dt}\left(\dfrac{\partial L}{\partial \dot{q}_i}\right) = \dfrac{\partial L}{\partial q_i}$	$\partial_\mu\left(\dfrac{\partial L}{\partial(\partial_\mu \phi_i)}\right) = \dfrac{\partial L}{\partial \phi_i}$

Key notational point

L is a Lagrangian *density* (energy per unit volume), not the total Lagrangian. The full Lagrangian is obtained by integrating: $L = \int L\, d^3x$. In relativistic field theory everyone writes L but calls it "the Lagrangian."

The Field–Theory Euler–Lagrange Equation

Master Equation for Fields

$$\boxed{\partial_\mu\left(\frac{\partial L}{\partial(\partial_\mu \phi_i)}\right) - \frac{\partial L}{\partial \phi_i} = 0}$$

Notice that the time derivative d/dt has been *promoted* to the four–divergence ∂_μ (a sum over $\mu = 0, 1, 2, 3$). This is the price of treating space and time on equal footing, as relativity demands.

Example 10.1: Klein–Gordon (Spin-0)

Term Refresher: Scalar Field

A **scalar field** $\phi(x, y, z, t)$ assigns a single real number to every point in spacetime (like temperature in a room). It describes **spin-0** particles (e.g. pions, the Higgs boson).

The Lagrangian is:

$$L = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}\left(\frac{mc}{\hbar}\right)^2 \phi^2 \tag{10.11}$$

What does this look like without the compact notation?

Writing out the Einstein summation explicitly (Griffiths uses metric $+, -, -, -$):

$$L = \frac{1}{2c^2}\left(\frac{\partial \phi}{\partial t}\right)^2 - \frac{1}{2}(\nabla \phi)^2 - \frac{1}{2}\left(\frac{mc}{\hbar}\right)^2 \phi^2$$

- First term $\sim$ "kinetic energy" (time variation of the field).
- Second term $\sim$ "elastic energy" (spatial variation / gradients).
- Third term $\sim$ "mass energy" (the field's rest energy cost).

This is the field–theory analogue of $L = \frac{1}{2}m\dot{q}^2 - V(q)$.

Applying the Euler–Lagrange equation to (10.11) yields the **Klein–Gordon equation**:

$$\partial_\mu \partial^\mu \phi + \left(\frac{mc}{\hbar}\right)^2 \phi = 0 \tag{10.13}$$

Written out component by component:

$$\frac{1}{c^2}\frac{\partial^2 \phi}{\partial t^2} - \nabla^2 \phi + \left(\frac{mc}{\hbar}\right)^2 \phi = 0$$

This is the relativistic wave equation for a spin-0 particle of mass m.

Example 10.2: Dirac (Spin-$\frac{1}{2}$)

Term Refresher: Spinor Field

A **spinor field** ψ is a four-component complex column vector at each spacetime point. It describes **spin-$\frac{1}{2}$** particles (electrons, quarks, neutrinos). The **adjoint spinor** is $\bar{\psi} \equiv \psi^\dagger \gamma^0$.

The Lagrangian:

$$L = i\hbar c\, \bar{\psi}\gamma^\mu \partial_\mu \psi - mc^2 \bar{\psi}\psi \tag{10.14}$$

Applying Euler–Lagrange (treating ψ and $\bar{\psi}$ as independent fields) yields the **Dirac equation**:

$$i\gamma^\mu \partial_\mu \psi - \frac{mc}{\hbar}\psi = 0 \tag{10.15}$$

Why two independent fields?

ψ is a complex 4-component spinor, so it contains *eight* real degrees of freedom. We could treat the eight real/imaginary parts separately, but it is much cleaner to treat the four components of ψ and the four components of $\bar{\psi}$ as independent. Applying Euler–Lagrange to $\bar{\psi}$ gives the Dirac equation; applying it to ψ gives the adjoint Dirac equation.

Example 10.3: Proca (Spin-1)

Term Refresher: Vector Field

A **vector field** A^μ assigns a four-vector to each spacetime point. It describes **spin-1** particles. Setting $m = 0$ gives the photon (electromagnetism); $m \neq 0$ gives a massive spin-1 boson (like the $W^\pm$ or Z^0).

The Lagrangian (with the shorthand $F^{\mu\nu} \equiv \partial^\mu A^\nu - \partial^\nu A^\mu$):

$$L = -\frac{1}{16\pi} F^{\mu\nu} F_{\mu\nu} + \frac{1}{8\pi}\left(\frac{mc}{\hbar}\right)^2 A^\nu A_\nu \tag{10.21}$$

Euler–Lagrange yields the **Proca equation**:

$$\partial_\mu F^{\mu\nu} + \left(\frac{mc}{\hbar}\right)^2 A^\nu = 0 \tag{10.22}$$

Connection to electrodynamics

Set $m = 0$ in (10.22) and you recover Maxwell's equations in empty space: $\partial_\mu F^{\mu\nu} = 0$. So electrodynamics is just the special case of a *massless* spin-1 field theory.

Example 10.4: Maxwell with Source

Adding a source current J^μ:

$$L = -\frac{1}{16\pi} F^{\mu\nu} F_{\mu\nu} - \frac{1}{c} J^\mu A_\mu \tag{10.23}$$

The Euler–Lagrange equation gives Maxwell's equations with sources:

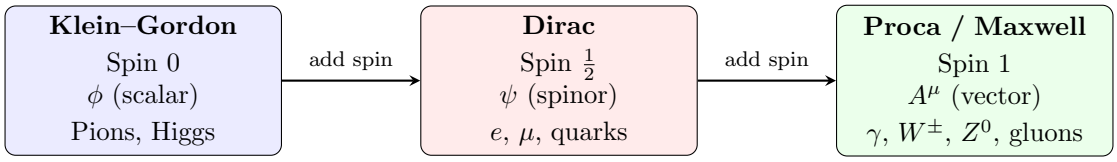
$$\partial_\mu F^{\mu\nu} = \frac{4\pi}{c} J^\nu \tag{10.24}$$

Built-in charge conservation

An automatic consequence of (10.24) is $\partial_\nu J^\nu = 0$ — the continuity equation. The Lagrangian framework *forces* charge conservation; you cannot put in an arbitrary current and still have consistent field equations.

The Big Picture: Three Lagrangians, Three Spins

Free–Field Lagrangians



These are FREE-field Lagrangians

Everything so far describes particles propagating *without interacting*. No forces, no scattering, no coupling. The entire point of Sections 10.3–10.4 is to show how interactions are **forced upon you** by requiring local gauge invariance.

3 Section 10.3 — Local Gauge Invariance

This is the heart of the chapter. The core question is:

The Central Question

What happens if we **demand** that the Lagrangian be invariant under a phase transformation $\psi \to e^{i\theta(x)}\psi$ where θ can vary from point to point?

Step 1: Global Gauge Invariance (the easy case)

Term Refresher: Global Gauge Transformation

A **global gauge transformation** (or global phase transformation) is:

$$\psi \;\to\; e^{i\theta}\,\psi \qquad (\theta = \text{constant, same everywhere})$$

Check: Is the Dirac Lagrangian $L = i\hbar c\,\bar{\psi}\gamma^\mu\partial_\mu\psi - mc^2\bar{\psi}\psi$ invariant?

Under $\psi \to e^{i\theta}\psi$, we get $\bar{\psi} \to e^{-i\theta}\bar{\psi}$. In every term of L, we find the combination $\bar{\psi}(\cdots)\psi$, and the two exponentials cancel: $e^{-i\theta}\cdot e^{i\theta} = 1$.

Result

The Dirac Lagrangian is invariant under global gauge transformations. ✓

This is unsurprising — in quantum mechanics, the overall phase of the wave function has never mattered.

Step 2: Local Gauge Invariance (the hard case)

Now let the phase depend on position:

$$\psi \;\to\; e^{i\theta(x)}\,\psi \qquad (\theta \text{ is a } \textit{function} \text{ of spacetime})$$

The mass term $\bar{\psi}\psi$ is still fine — the exponentials still cancel. But the derivative term $\bar{\psi}\gamma^\mu\partial_\mu\psi$ is **not** invariant, because ∂_μ acts on the phase too:

$$\partial_\mu\psi \;\to\; \partial_\mu\big(e^{i\theta(x)}\psi\big) = e^{i\theta}\big(\partial_\mu\psi + i(\partial_\mu\theta)\,\psi\big) \tag{10.28}$$

The extra term $i(\partial_\mu\theta)\psi$ spoils the invariance. In fact:

$$L \;\to\; L \;-\; \hbar c\,(\partial_\mu\theta)\,\bar{\psi}\gamma^\mu\psi \tag{10.29}$$

Physical picture

If you rotate the phase by different amounts at different points, derivatives "notice" the mismatch. It's like having a ruler whose zero-mark shifts from place to place — you can't take a consistent derivative anymore.

Step 3: The Fix — Introduce a Gauge Field

The Covariant Derivative

Replace every ordinary derivative ∂_μ with a **covariant derivative**:

$$D_\mu \;\equiv\; \partial_\mu + \frac{iq}{\hbar c}\,A_\mu$$

where A_μ is a brand–new vector field (the **gauge field**), and q is the charge of the particle.

For this to work, we also need A_μ to transform in a specific way:

$$A_\mu \;\to\; A_\mu + \partial_\mu\lambda \tag{10.34}$$

where $\lambda(x) = -(\hbar c/q)\,\theta(x)$.

When A_μ transforms according to (10.34), the ugly $\partial_\mu\theta$ terms *cancel*, and the covariant derivative transforms cleanly:

$$D_\mu\psi \;\to\; e^{i\theta(x)}\,D_\mu\psi$$

— exactly the same way ψ itself transforms. So $\bar{\psi}\gamma^\mu D_\mu\psi$ is invariant. ✓

Step 4: The Gauge Field Needs Its Own Lagrangian

We've introduced a new field A_μ. It needs a kinetic ("free") term of its own. We want this term to also be gauge-invariant.

The combination $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ *is* invariant under $A_\mu \to A_\mu + \partial_\mu \lambda$ (the extra $\partial_\mu \partial_\nu \lambda$ terms cancel by symmetry of mixed partials).

However, a mass term $A_\mu A^\mu$ is **not** invariant. This forces the gauge field to be **massless**.

The Complete QED Lagrangian

$$\boxed{L = \underbrace{i\hbar c\,\bar{\psi}\gamma^\mu \partial_\mu \psi - mc^2 \bar{\psi}\psi}_{\text{free Dirac}} - \underbrace{\frac{1}{16\pi} F^{\mu\nu} F_{\mu\nu}}_{\text{free Maxwell}} - \underbrace{q\bar{\psi}\gamma^\mu \psi\, A_\mu}_{\text{interaction}}} \tag{10.35}$$

What just happened?

Starting from the free Dirac Lagrangian and *demanding* local $U(1)$ gauge invariance:

1. We were **forced** to introduce a massless vector field A_μ.
2. The coupling $q\bar{\psi}\gamma^\mu\psi\, A_\mu$ was **uniquely determined**.
3. Adding the kinetic term $F_{\mu\nu}F^{\mu\nu}$ gave us **Maxwell's equations** with the Dirac current $J^\mu = cq\bar{\psi}\gamma^\mu\psi$ as source.

All of electrodynamics emerges from a symmetry requirement. The photon's existence is not an input — it is a *consequence*.

The Minimal Coupling Rule

The entire procedure boils down to a simple recipe:

$$\partial_\mu \;\longrightarrow\; D_\mu = \partial_\mu + \frac{iq}{\hbar c} A_\mu$$

Take any free Lagrangian, replace all derivatives with covariant derivatives, add the free Lagrangian for the gauge field, and you automatically get the correct interacting theory. This is called the **minimal coupling rule**.

Why "gauge"?

The name comes from Hermann Weyl's 1918 attempt to unify gravity and electromagnetism using scale ("gauge") transformations. That particular idea didn't work, but the name stuck. Today "gauge transformation" just means a local phase transformation of the fields.

4 Section 10.4 — Yang–Mills Theory

The Upgrade

Section 10.3 used a $U(1)$ phase: $\psi \to e^{i\theta}\psi$.
Now we ask: what if the symmetry group is **larger**? Specifically, what about $SU(2)$?

The Setup: Two Degenerate Dirac Fields

Suppose we have two spin-$\frac{1}{2}$ fields ψ_1 and ψ_2 with the *same mass* m. We group them into a column:

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} \tag{10.42}$$

The free Lagrangian:

$$L = i\hbar c\, \bar{\psi}\gamma^\mu \partial_\mu \psi - mc^2 \bar{\psi}\psi \tag{10.46}$$

which is shorthand for the sum of the individual Dirac Lagrangians.

Term Refresher: $SU(2)$

$SU(2)$ is the group of 2×2 unitary matrices with determinant 1. Every element can be written as $S = e^{i\tau\cdot\boldsymbol{\theta}/2}$ where $\tau = (\tau_1, \tau_2, \tau_3)$ are the Pauli matrices and $\boldsymbol{\theta}$ is a vector of three parameters.

You already know this group from angular momentum / spin-$\frac{1}{2}$ in quantum mechanics! $SU(2)$ acts on the two-component object ψ the same way spin rotations act on a two-component spinor.

Global $SU(2)$ Invariance

The free Lagrangian (10.46) is invariant under:

$$\psi \;\to\; S\,\psi \qquad \text{where } S \text{ is a } \textit{constant } SU(2) \text{ matrix}$$

because $\bar{\psi} \to \bar{\psi}S^{-1}$, and $S^{-1}S = \mathbf{1}$.

Local $SU(2)$ Invariance

Now promote S to $S(x)$ — a different $SU(2)$ rotation at each spacetime point:

$$\psi \;\to\; S(x)\,\psi, \qquad S(x) = \exp\!\left(-\frac{iq}{\hbar c}\,\tau\cdot\boldsymbol{\lambda}(x)\right)$$

The derivative term again breaks: $\partial_\mu\psi \to S(\partial_\mu\psi) + (\partial_\mu S)\psi$, and the extra piece $(\partial_\mu S)\psi$ spoils invariance.

The Fix: Three Gauge Fields

To restore invariance, define a covariant derivative:

$$D_\mu = \partial_\mu + \frac{iq}{\hbar c}\,\tau \cdot \mathbf{A}_\mu$$

where $\mathbf{A}_\mu = (A_\mu^1,\, A_\mu^2,\, A_\mu^3)$ are **three** new gauge fields — one for each generator of $SU(2)$ (i.e. one for each Pauli matrix).

Counting Gauge Fields

The number of gauge fields always equals the number of generators of the symmetry group:

Group	Generators	Gauge Fields
$U(1)$	1	1 (the photon)
$SU(2)$	3	3 (W^+, W^-, Z^0 in electroweak)
$SU(3)$	8	8 (the gluons)

Two New Complications (compared to $U(1)$)

1. **The gauge field transformation is more complicated.**

   In the $U(1)$ case: $A_\mu \to A_\mu + \partial_\mu \lambda$.

   In $SU(2)$, to first order in $\boldsymbol{\lambda}$:

$$\mathbf{A}_\mu \;\to\; \mathbf{A}_\mu + \partial_\mu \boldsymbol{\lambda} + \frac{2q}{\hbar c}\,(\boldsymbol{\lambda} \times \mathbf{A}_\mu) \tag{10.61}$$

   The cross product term is *new* — it arises because $SU(2)$ is **non-Abelian** (its generators don't commute: $[\tau_i, \tau_j] = 2i\epsilon_{ijk}\tau_k$).

2. **The field strength tensor is modified.**

   In $U(1)$: $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

   In $SU(2)$:

$$\mathbf{F}_{\mu\nu} = \partial_\mu \mathbf{A}_\nu - \partial_\nu \mathbf{A}_\mu - \frac{2q}{\hbar c}\,(\mathbf{A}_\mu \times \mathbf{A}_\nu) \tag{10.65}$$

   The cross product term means the field strength depends *nonlinearly* on the gauge fields — in physical terms, the gauge bosons **carry charge** and interact with each other. Photons don't self-interact; gluons do.

The Complete Yang–Mills Lagrangian

Yang–Mills Lagrangian

$$\boxed{L = \left[i\hbar c\,\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi - mc^{2}\bar{\psi}\psi\right] - \frac{1}{16\pi}\mathbf{F}^{\mu\nu}\cdot\mathbf{F}_{\mu\nu} - q\left(\bar{\psi}\gamma^{\mu}\tau\psi\right)\cdot\mathbf{A}_{\mu}} \tag{10.67}$$

This describes two equal-mass Dirac fields interacting with three massless vector gauge fields.

Self–interaction of gauge fields

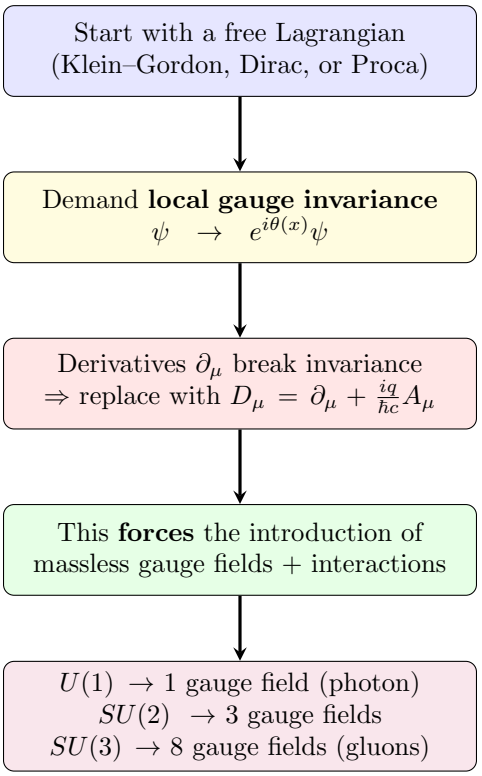
Because $\mathbf{F}_{\mu\nu}$ contains the cross-product term $\mathbf{A}_{\mu}\times\mathbf{A}_{\nu}$, the "free" gauge Lagrangian $\mathbf{F}^{\mu\nu}\cdot\mathbf{F}_{\mu\nu}$ already contains **cubic and quartic** terms in $\mathbf{A}_{\mu}$. These correspond to three-boson and four-boson vertices — the gauge bosons interact with themselves!

This does *not* happen in QED, where $F_{\mu\nu}$ is linear in A_{μ}, and photon-photon vertices are absent at tree level. It *does* happen in QCD (gluon self-coupling) and in the electroweak theory (W/Z self-coupling).

The mass problem

Local gauge invariance again forces all three gauge fields to be **massless** (the Proca mass term is not gauge-invariant). But the $W^{\pm}$ and Z^{0} bosons are *massive* ($\sim$ 80–91 GeV). This is a serious problem — it will be resolved later in the chapter by the **Higgs mechanism** (Section 10.9), which gives gauge bosons mass without breaking gauge invariance of the underlying Lagrangian.

Master Summary: The Logic of Sections 10.1–10.4



The Big Takeaway

Symmetry dictates interaction. You do not put forces in by hand. You start with free particles, impose a symmetry requirement (local gauge invariance), and the forces — complete with their gauge bosons, coupling rules, and conservation laws — emerge automatically. This is the conceptual foundation of the entire Standard Model.